

Hackathon Problem Statements

Regd No: 25091A3270, 25091A3271

Problem Statement No: 14

Post-Harvest Storage Loss Predictor Build a predictive model that estimates storage loss risk for different crops based on humidity, temperature, and pest presence data in warehouses, using classification algorithms trained on historical spoilage records. Alert warehouse managers via a priority-queue-based ranking system when intervention is urgently needed for specific storage units. Design storage units and crop batches as OOP objects with shelf-life attributes. This directly reduces the massive food wastage occurring in unregulated rural storage.

Regd No: 25091A32E1, 25091A32P3

Problem Statement No: 18

Blood Bank Inventory & Donor Matching System Create a blood bank management system using hash maps for $O(1)$ blood-type lookups and OOP inheritance to model different donor categories (regular, emergency, first-time). Apply time-series forecasting to predict blood shortage periods based on historical donation and demand patterns. The system should auto-notify compatible donors during critical shortages using a matching algorithm. This addresses frequent blood shortages faced by rural and semi-urban blood banks.

Regd No: 25091A3259, 25091A3254

Problem Statement No: 61

Skill-Gap Analyzer Design a skill-gap analysis tool that clusters student skill profiles (from resumes/self-assessments) and compares them against current job market demand data scraped from public job portals, using classification to identify missing in-demand skills. Use OOP to model students, skills, and job requirements. This helps career counseling cells guide students toward targeted upskilling before placements.

Regd No: 25091A3283, 25091A32C2

Problem Statement No: 71

Deforestation Rate Tracker Build a deforestation trend tracker using time-series regression on publicly available forest cover datasets to estimate rate of loss by region and project future cover under current trends. Visualize year-over-year change to support conservation policy advocacy. Use OOP to structure regions, datasets, and analysis pipelines. This provides evidence-based insight for forest departments and environmental NGOs working on conservation planning.

Regd No: 25091A3228, 25091A3227

Problem Statement No: 93

Rural Banking Transaction Fraud Detector Build a fraud detection system using anomaly detection algorithms on simulated banking transaction datasets to flag unusual patterns (sudden large withdrawals, atypical transaction locations/times) common in rural banking fraud cases. Use OOP to model accounts, transactions, and fraud alerts. This helps protect vulnerable, often less digitally-literate rural banking customers from financial fraud.

Regd No: 25091A32M7, 25091A3222

Problem Statement No: 112

Missing Person Search-and-Track System Build a missing person tracking system using graph-based last-seen network analysis to model reported sighting locations as connected nodes, helping investigators identify likely search zones based on movement patterns. Use OOP to model persons, sightings, and search zones. This supports police and disaster response teams in systematically organizing search efforts rather than relying on ad-hoc methods.